

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-3342
Project Name	AI BlogNest API
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	No frontend client UI – API-only backend (Client/ folder reserved but empty)	Harder for non-technical evaluators/users to try the product directly	Medium
2	Single local MongoDB instance, not yet migrated to a managed cloud cluster	No built-in redundancy if the local database instance goes down	Medium
3	POST /api/ai/generate-blog can take up to ~38 seconds due to Gemini's generation latency	Slower perceived responsiveness for the AI Blog Generator feature specifically	High
4	No dedicated Categories collection or likes/reactions layer yet	Category browsing and engagement metrics are limited to a plain string field and a simple counter	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local Node.js instance	Deploy stateless instances behind a load balancer – JWT auth requires no server-side session, so horizontal scaling is straightforward
2	Data Storage	Single local MongoDB instance	Migrate to a managed MongoDB cluster with replica sets, and add a dedicated Categories collection
3	AI Performance	Synchronous Gemini call blocks the request for up to ~38 seconds	Move AI Blog Generation to an async job queue with a webhook/poll pattern so the client is not left waiting on one open connection
4	Security	JWT + role middleware + input validation	Add rate-limiting middleware, refresh tokens, and helmet.js security headers

Future Roadmap

The extended data model explored during design (see the ER Diagram in Project Documentation) already outlines the next three phases of growth:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add a dedicated Category collection (name, slug, description) instead of a plain string field, plus a Likes relation between Users and Blog posts	October 2026	Better content discovery and genuine engagement metrics
Phase 3	Enrich the Blog schema with slug, excerpt and cover image fields to match the extended ER design	November 2026	Richer post previews and shareable, SEO-friendly URLs
Phase 4	Build the reserved Client/ folder into a lightweight frontend, plus an admin analytics dashboard over usage and moderation logs	December 2026	Direct end-user access and operational visibility beyond API clients